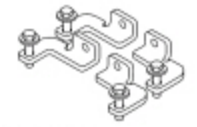
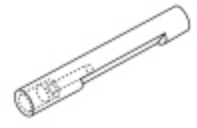


Valve Spring, Retainer and Seal



Special Tool(s)

 ST1981-A	Compressor, Valve Spring 303-300 (T87C-6565-A)
 ST1907-A	Compressor, Valve Spring 303-350 (T89P-6565-A)
 ST3028-A	Compressor, Valve Spring 303-1249
 ST1906-A	Installer, Valve Stem Oil Seal 303-470 (T94P-6510-CH)

Material

Item	Specification
Motorcraft® SAE 5W-20 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

Removal

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

1. Remove the valve tappets from the cylinder being serviced. For additional information, refer to Valve Tappets in this section.
2. Rotate the crankshaft until the piston for the valve being serviced is at the top of its stroke.

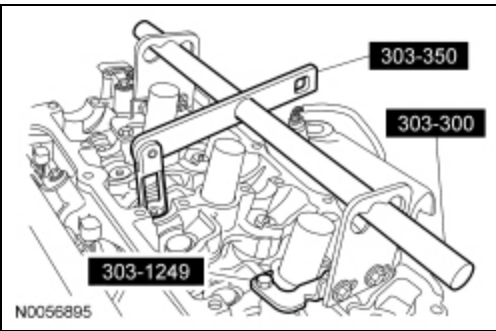
3. **NOTICE:** If air pressure has forced the piston to the bottom of the cylinder, any loss of air pressure will allow the valve to fall into the cylinder. If air pressure must be removed, support the valve prior to removal or engine damage may occur.

NOTE: If the components are to be reinstalled, they must be installed in the same positions. Mark the components for installation into their original locations.

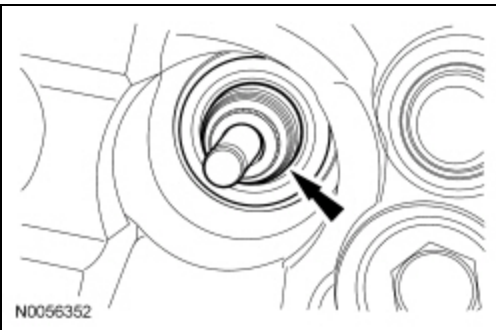
NOTE: If a valve drops into the cylinder, remove the cylinder head. For additional information, refer to Cylinder Head — RH or Cylinder Head — LH in this section.

Pressurize the cylinder using compressed air.

4. Using the Valve Spring Compressors, remove the keys, retainer and spring.



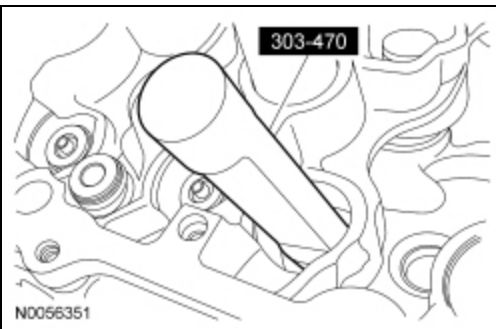
5. Remove and discard the valve stem seal.



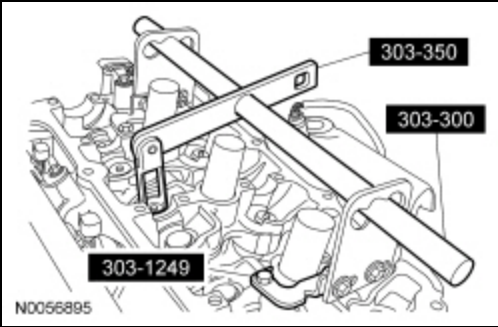
Installation

1. **NOTE:** Lubricate the valve stem seal with clean engine oil prior to installation.

Using the Valve Stem Oil Seal Installer, install a new valve stem seal.



2. Using the Valve Spring Compressors, install the valve spring, retainer and key.



3. Install the valve tappets. For additional information, refer to Valve Tappets in this section.